Beltrami PPE Swabs

Methodology

PPE swabs were run by RT-QuIC at 42°C for 72 hours. Maxpoint ratios and max slope values were used as predictors in a generalized linear model for the following factors:

1. Location (dump site, negative site, perimeter)
2. Timing (pre/post-sampling and post-decon)
3. Surface (the surface that was swabbed).

Least-square means were calculated, and pairwise comparisons were performed for each factor. Contrasts with a p-value ≤0.05 were considered significant.

Results

Table 1: Maxpoint ratio results. Significant contrasts are highlighted in red.

Factor	Contrast	Estimate	Std.error	p.value	Metric	Result
Location	negative site - perimeter	-1.198	0.2854	9.377e-05	MPR	***
Location	negative site - dump site	-0.768	0.2493	0.006127	MPR	**
Location	dump site - perimeter	-0.4296	0.1866	0.0564	MPR	ns
Timing	pre sampling - post sampling	-0.2581	0.15	0.08581	MPR	ns
Surface	tyvek - boot	-1.211	0.1834	9.942e-10	MPR	***
Surface	blank - boot	-1.522	0.2459	7.532e-09	MPR	***
Surface	boot - plastic_boot	0.6559	0.2005	0.006219	MPR	**
Surface	blank - plastic_boot	-0.8658	0.2722	0.008401	MPR	**
Surface	tyvek - plastic_boot	-0.5554	0.2192	0.05597	MPR	ns
Surface	tyvek - blank	0.3104	0.2614	0.635	MPR	ns

Table 2: Max slope results. Significant contrasts are highlighted in red.

Factor	Contrast	Estimate	Std.error	p.value	Metric	Result
Location	negative site - perimeter	-0.4147	0.1013	0.0001449	MS	***
Location	negative site - dump site	-0.3111	0.08849	0.001377	MS	**
Location	dump site - perimeter	-0.1036	0.06626	0.2626	MS	ns
Timing	pre sampling - post sampling	-0.09028	0.05324	0.09051	MS	ns
Surface	tyvek - boot	-0.3705	0.06512	1.242e-07	MS	***
Surface	blank - boot	-0.4204	0.0873	1.126e-05	MS	***
Surface	boot - plastic_boot	0.199	0.07116	0.02739	MS	*
Surface	blank - plastic_boot	-0.2214	0.09662	0.1011	MS	ns
Surface	tyvek - plastic_boot	-0.1715	0.07783	0.1234	MS	ns
Surface	tyvek - blank	0.04993	0.0928	0.9497	MS	ns

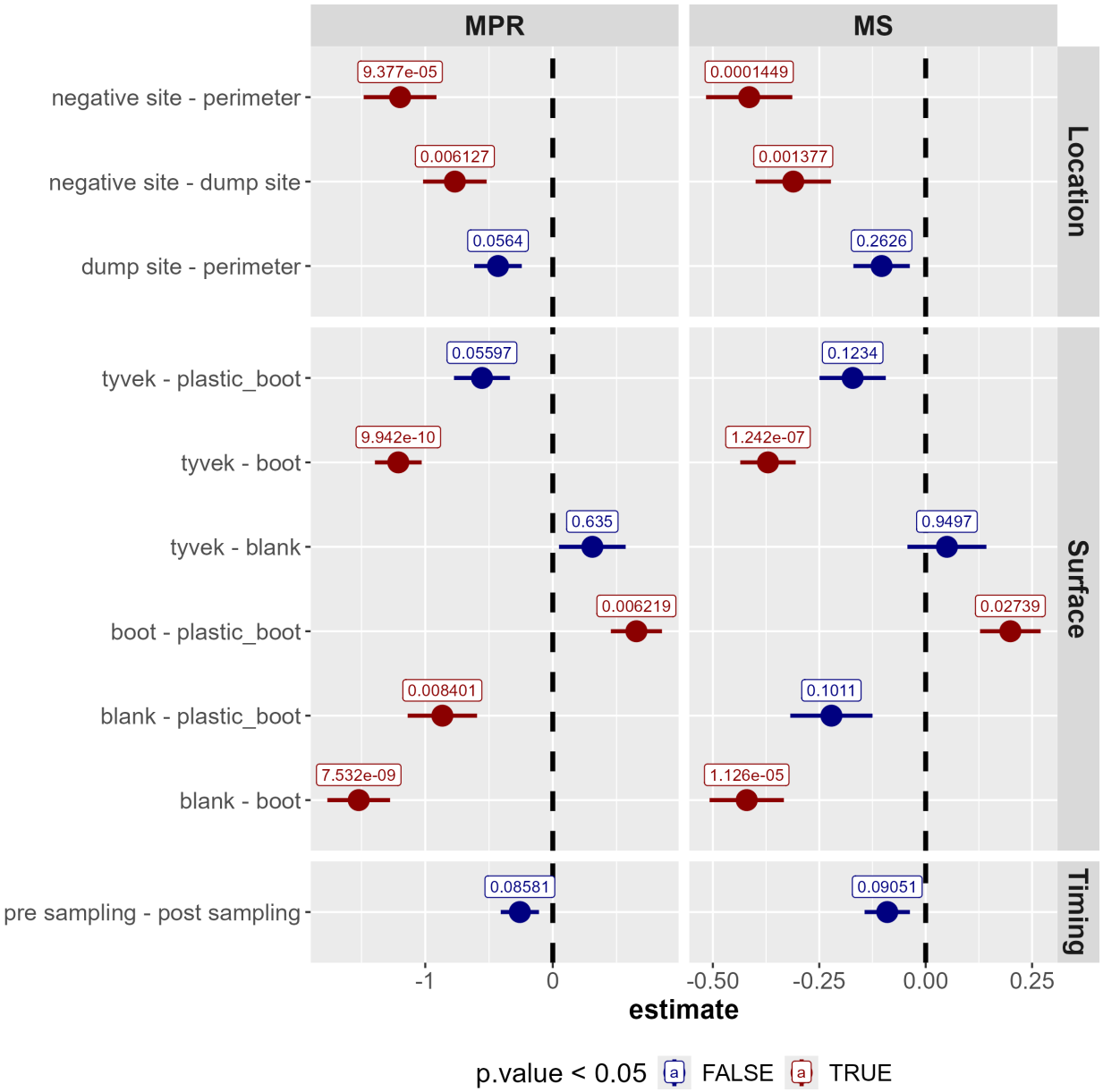


Figure 1: Estimates of each contrast. Plot is facetted by factor and metric (MPR or MS). Estimates are labelled with their p-value and colored based on significance.

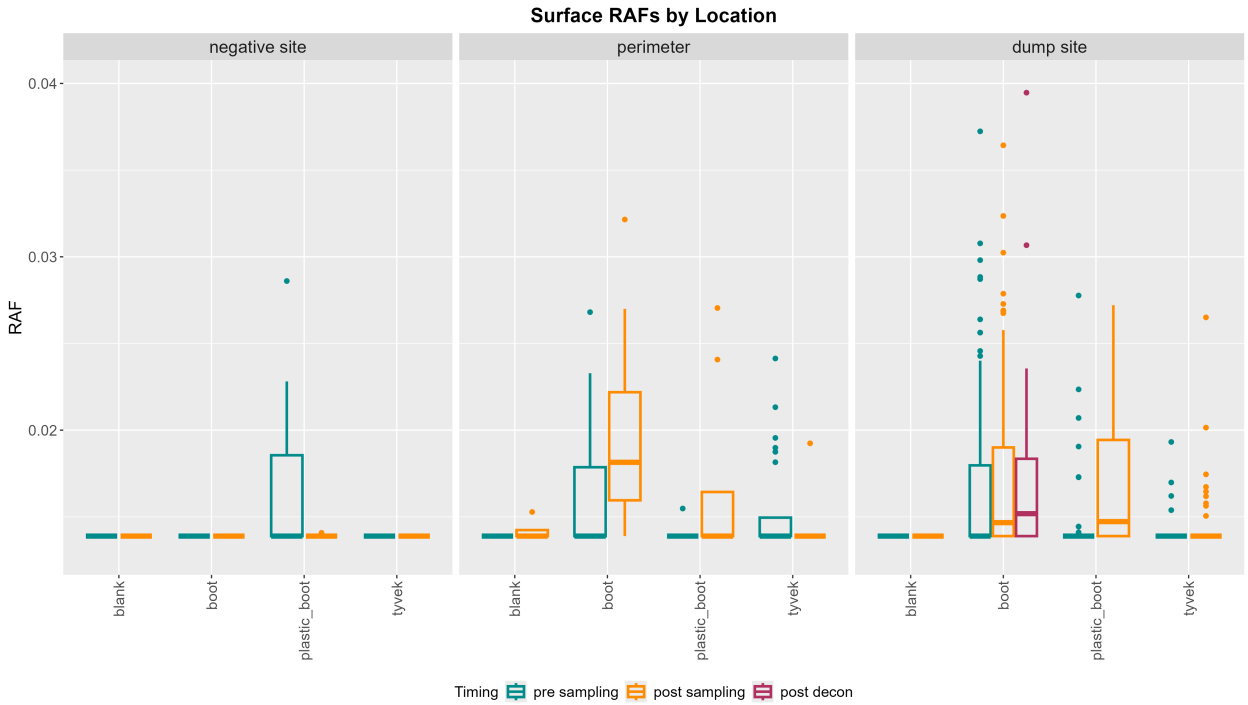


Figure 2: Rates of amyloid formation by swab surface. Figure is facetted based on location and colored based on timing.

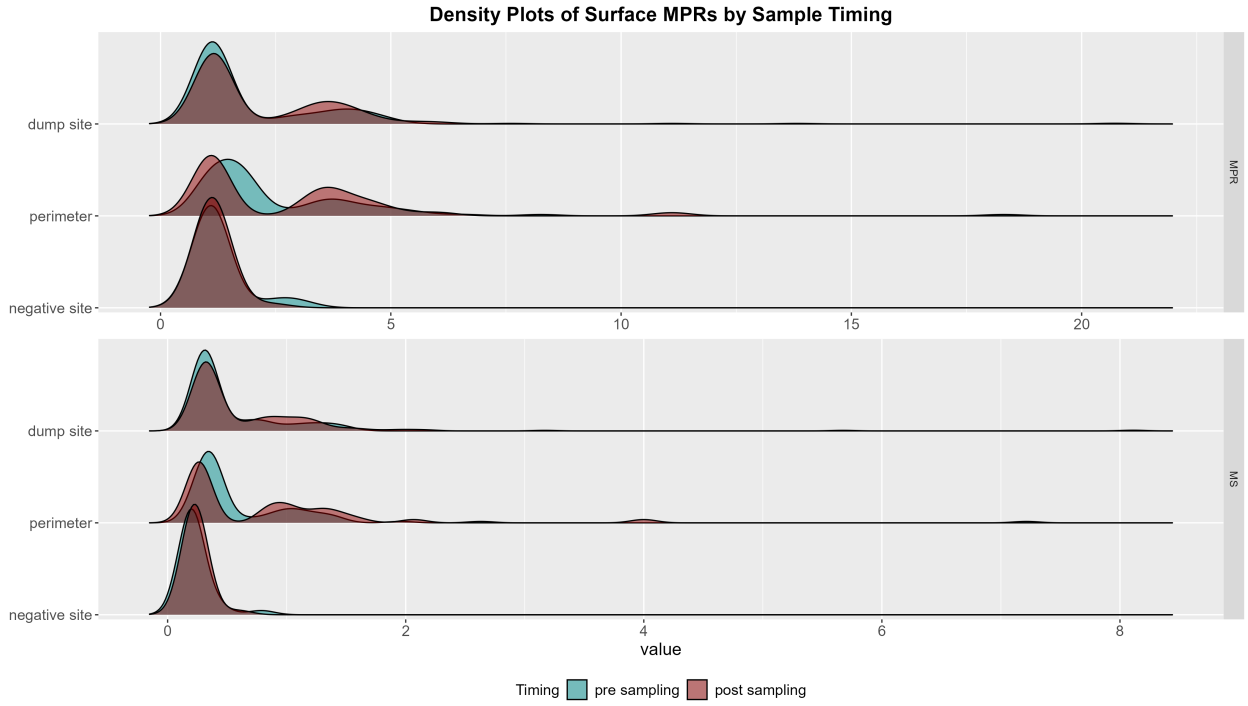


Figure 3: Density plot of MPRs for each location. Curves are filled based on timing.